Re3900 pressure coupling: development verification

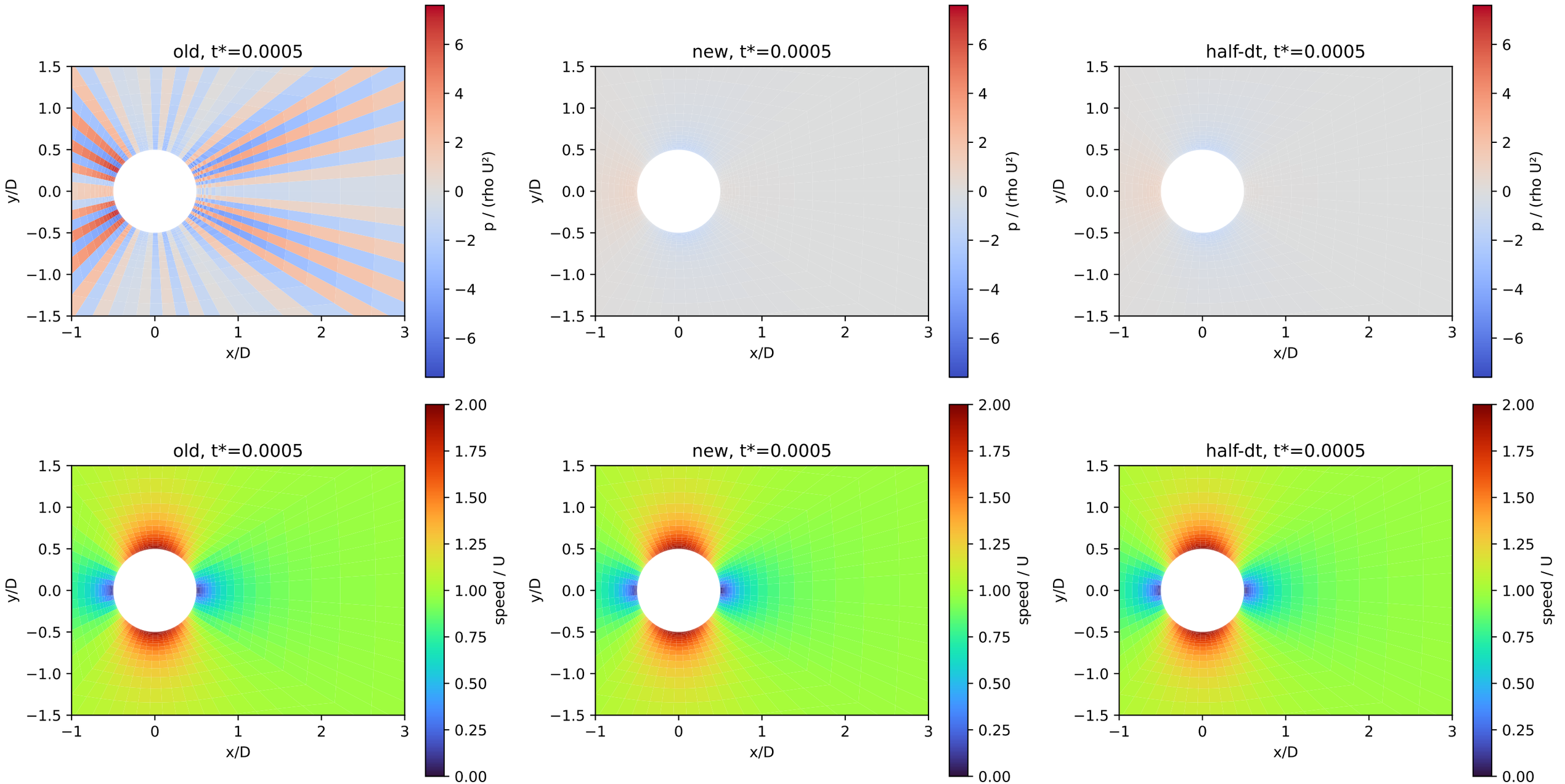
Matched startup: 24,576 native body-fitted cells, tU/D=0.0005

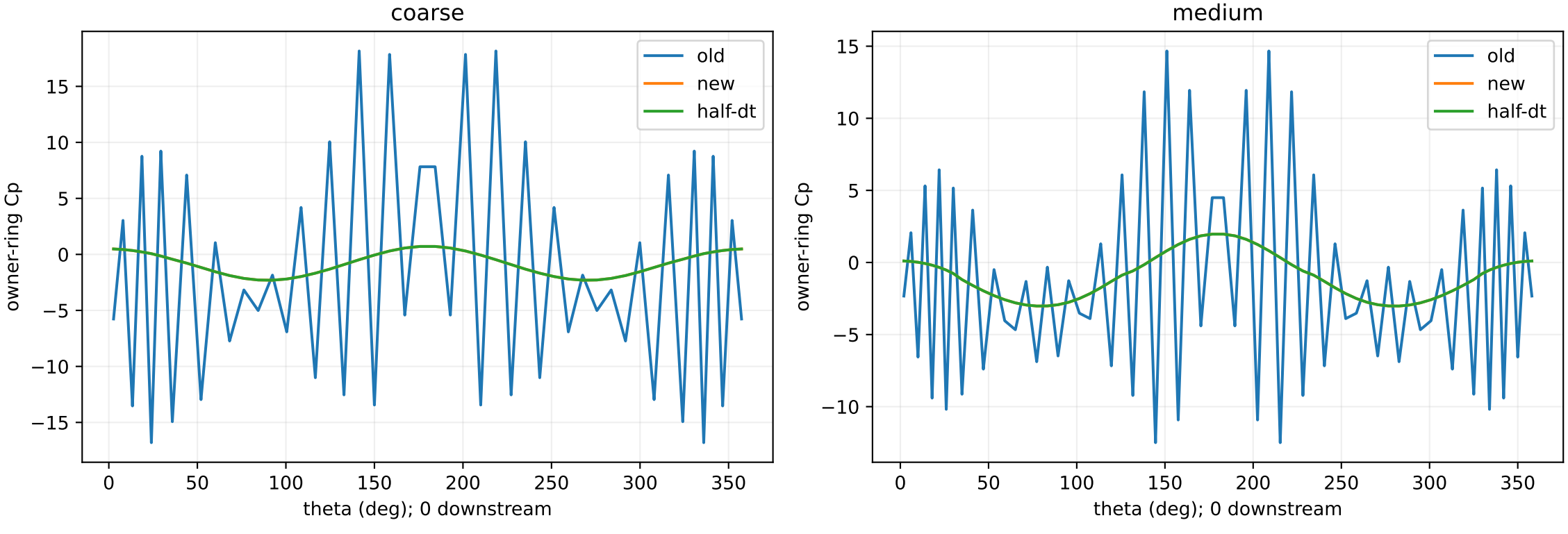
Old high-angular-frequency pressure fraction: 93.232%
New fraction: 0.00733%
Pressure L2 dt versus dt/2: 0.0627%

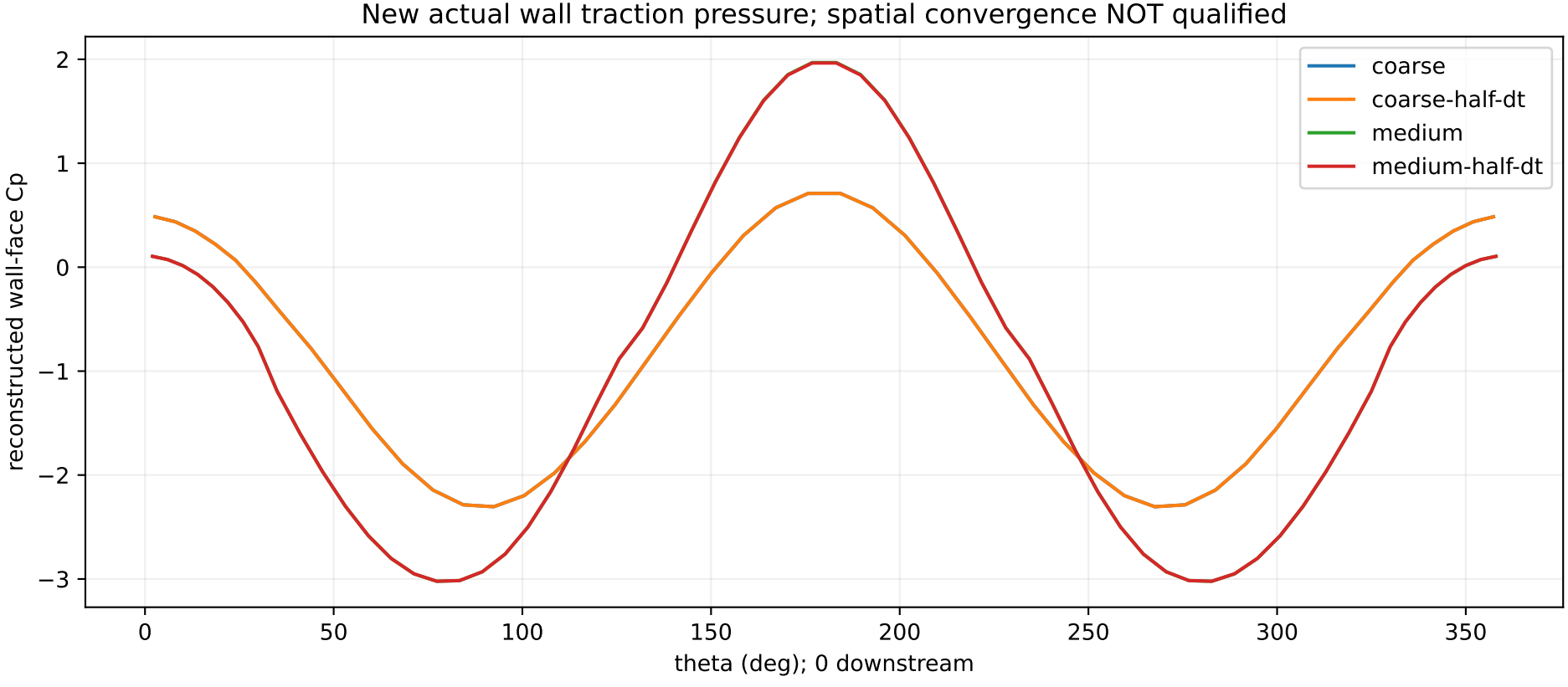
Changes: retained native face-flux history, boundary-consistent momentum increments, conservative skew-correct face pressure.

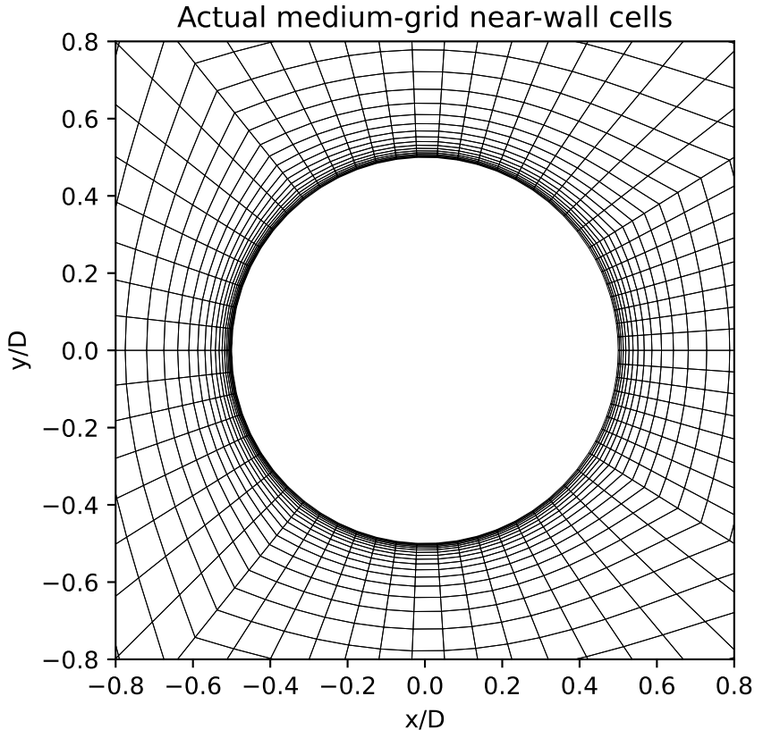
Remaining cell/face normal mismatch: 0.239 U
Interpolated cell-velocity divergence: 123 U/D
Native face divergence: 8.57e-09 U/D

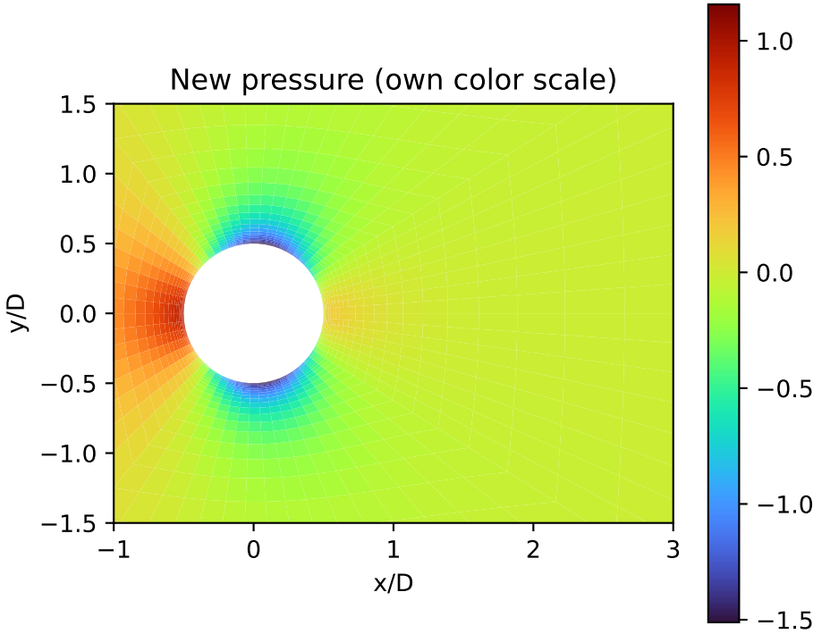
NUMERICAL / PHYSICAL QUALIFICATION: FALSE
Short-time pressure improvement is not verified Re3900 LES.
Cell/face compatibility, grid refinement and long-time statistics remain required. See report.md and comparison.json for full evidence.

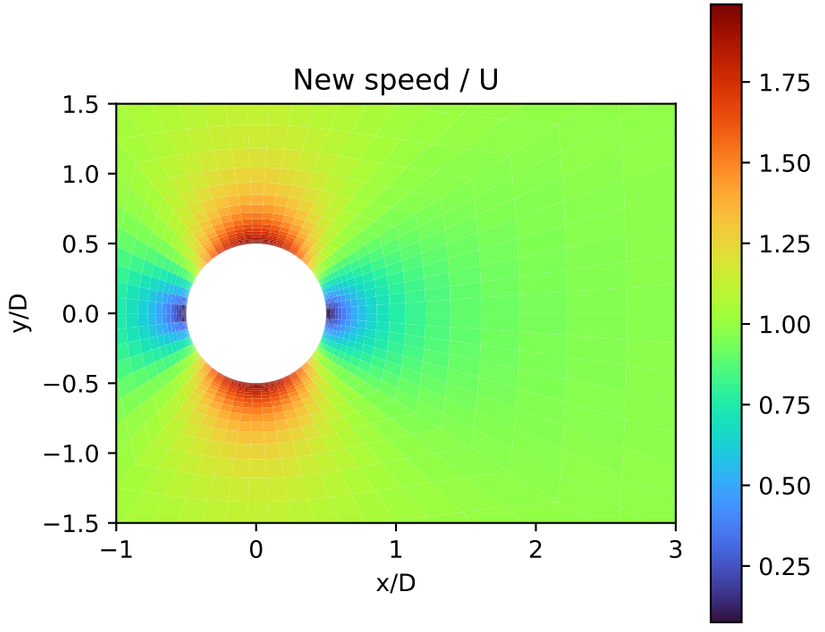

Unsmoothed native-cell fields: identical pressure / speed color scales
old, t*=0.0005
new, t*=0.0005
half-dt, t*=0.0005
x/D
y/D
p / (rho U²)
speed / U


Identical adjacent-cell definition for old/new comparison; no experiment implied
coarse
medium
old
new
half-dt
owner-ring Cp
theta (deg); 0 downstream
15
10
5
0
−5
−10
−15
0
50
100
150
200
250
300
350


New actual wall traction pressure; spatial convergence NOT qualified
coarse
coarse-half-dt
medium
medium-half-dt
reconstructed wall-face Cp
theta (deg); 0 downstream
2
1
0
−1
−2
−3
0
50
100
150
200
250
300
350


Actual medium-grid near-wall cells
y/D
x/D
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−0.8 −0.6 −0.4 −0.2 0.0 0.2 0.4 0.6 0.8


New pressure (own color scale)
y/D
x/D
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−1 0 1 2 3
1.0
0.5
0.0
−0.5
−1.0
−1.5


New speed / U
y/D
x/D
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−1 0 1 2 3
1.75
1.50
1.25
1.00
0.75
0.50
0.25